

Chapter 1 Overview

I Objective of Computer security

Three key objectives:

- Integrity : guarding against improper information modification or destruction

Examples: banking information, medical records

- Confidentiality : private information is not made available or disclosed to unauthorized individuals

Examples: SMS, password, personally identifiable information

- Availability : information delivery is not denied to authorized users

Examples: firewall should not stop running, internet access
military communication

II Security attacks

Any action that compromises the security of information

II. A Passive attacks

The goal of the adversary is to obtain information about confidential data without interfering with the communication.

Examples: eavesdropping, monitoring transmissions,

Remark: wireless communication is prone to eavesdropping due to the broadcast nature of the transmission medium.

Remark: passive attacks are, in general, difficult to detect. Prevention (with encryption of information) is easier than detection.

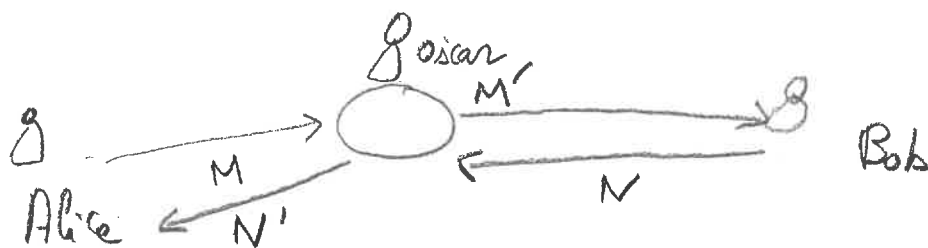
II. B Active attacks

Involve some modification of the data stream or the creation of a false stream of information.

Examples:

- Denial of service
- Signal jamming
- Ransom attack
- Impersonation: one entity pretends to be a different entity

- Man-in-the-middle attack



Remark: In general, active attacks are more difficult to combat than passive attacks.
 Countless new and evolving attacks.

III Security mechanism

A process that is designed to detect, prevent, or recover from a security attack.

Examples:

- firewalls
- authentication / access control
- backup
- counterbacking
- encryption
- honeypot
- VPN